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import tkinter as tk
from tkinter import messagebox
from time import strftime
import winsound
import threading

class DigitalClockAlarm:
    def __init__(self, root):
        self.root = root
        self.root.title("Digital Clock with Alarm")
        self.root.geometry("450x350")
        self.root.resizable(False, False)
        self.root.configure(bg='black')

        self.alarm_time = None
        self.alarm_active = False

        # Time display
        self.time_label = tk.Label(
            root,
            font=('Arial', 48, 'bold'),
            background='black',
            foreground='cyan'
        )
        self.time_label.pack(pady=10)

        # Date display
        self.date_label = tk.Label(
            root,
            font=('Arial', 14),
            background='black',
            foreground='white'
        )
        self.date_label.pack()

        # Alarm section
        alarm_frame = tk.Frame(root, bg='black')
        alarm_frame.pack(pady=20)

        tk.Label(
            alarm_frame,
            text="Set Alarm (HH:MM:SS):",
            font=('Arial', 12),
            bg='black',
            fg='white'
        ).grid(row=0, column=0, columnspan=3, pady=5)

        # Hour input
        self.hour_var = tk.StringVar(value="00")
        self.hour_entry = tk.Entry(
            alarm_frame,
            textvariable=self.hour_var,
            width=5,
            font=('Arial', 16),
            justify='center'
        )
        self.hour_entry.grid(row=1, column=0, padx=2)

        tk.Label(alarm_frame, text=":", font=('Arial', 16), bg='black', fg='white').grid(row=1,
column=1)

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# Minute input
self.minute_var = tk.StringVar(value="00")
self.minute_entry = tk.Entry(
    alarm_frame,
    textvariable=self.minute_var,
    width=5,
    font=('Arial', 16),
    justify='center'
)
self.minute_entry.grid(row=1, column=2, padx=2)

tk.Label(alarm_frame, text=":", font=('Arial', 16), bg='black', fg='white').grid(row=1,
column=3)

# Second input
self.second_var = tk.StringVar(value="00")
self.second_entry = tk.Entry(
    alarm_frame,
    textvariable=self.second_var,
    width=5,
    font=('Arial', 16),
    justify='center'
)
self.second_entry.grid(row=1, column=4, padx=2)

# Buttons
button_frame = tk.Frame(root, bg='black')
button_frame.pack(pady=10)

self.set_button = tk.Button(
    button_frame,
    text="Set Alarm",
    font=('Arial', 12),
    bg='green',
    fg='white',
    command=self.set_alarm,
    width=12
)
self.set_button.grid(row=0, column=0, padx=5)

self.cancel_button = tk.Button(
    button_frame,
    text="Cancel Alarm",
    font=('Arial', 12),
    bg='red',
    fg='white',
    command=self.cancel_alarm,
    width=12,
    state='disabled'
)
self.cancel_button.grid(row=0, column=1, padx=5)

# Alarm status
self.status_label = tk.Label(
    root,
    text="No alarm set",
    font=('Arial', 11),
    bg='black',
    fg='yellow'

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)
self.status_label.pack(pady=5)

# Start clock
self.update_time()

def update_time(self):
    time_string = strftime('%H:%M:%S')
    date_string = strftime('%A, %B %d, %Y')

    self.time_label.config(text=time_string)
    self.date_label.config(text=date_string)

    # Check alarm
    if self.alarm_active and time_string == self.alarm_time:
        self.trigger_alarm()

    self.time_label.after(1000, self.update_time)

def set_alarm(self):
    try:
        hour = int(self.hour_var.get())
        minute = int(self.minute_var.get())
        second = int(self.second_var.get())

        if not (0 <= hour <= 23 and 0 <= minute <= 59 and 0 <= second <= 59):
            messagebox.showerror("Invalid Time", "Please enter valid time values!")
            return

        self.alarm_time = f"{hour:02d}:{minute:02d}:{second:02d}"
        self.alarm_active = True

        self.status_label.config(text=f"Alarm set for: {self.alarm_time}", fg='lime')
        self.set_button.config(state='disabled')
        self.cancel_button.config(state='normal')

        messagebox.showinfo("Alarm Set", f"Alarm set for {self.alarm_time}")

    except ValueError:
        messagebox.showerror("Invalid Input", "Please enter numeric values!")

def cancel_alarm(self):
    self.alarm_active = False
    self.alarm_time = None

    self.status_label.config(text="No alarm set", fg='yellow')
    self.set_button.config(state='normal')
    self.cancel_button.config(state='disabled')

    messagebox.showinfo("Alarm Cancelled", "Alarm has been cancelled")

def trigger_alarm(self):
    self.alarm_active = False

    def play_sound():
        try:
            # Play system beep sound (works on Windows)
            for _ in range(5):
                winsound.Beep(1000, 500)
        except:

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# Fallback for other systems
print('\a' * 5)

# Play sound in separate thread
threading.Thread(target=play_sound, daemon=True).start()

messagebox.showwarning("ALARM!", f"Alarm ringing!\nTime: {self.alarm_time}")

self.status_label.config(text="Alarm triggered!", fg='red')
self.set_button.config(state='normal')
self.cancel_button.config(state='disabled')

# Create and run application
root = tk.Tk()
app = DigitalClockAlarm(root)
root.mainloop()
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